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Pre-analysis:
For the pre-analysis, it is important to clean/filter the data to fit what we require. First of all, anyone who is not an undergraduate at UVA needs to be filtered out. Furthermore, anyone who has “never” had a hotdog should be filtered out as well. A tool we can possibly use here includes Pandas. We can conduct box and whisker plots to make sure there are not any major outliers in the data to ensure there is no skew.

Analysis:
In order to test the hypothesis, a two-tailed t-test must be conducted. This is because we are testing for a general direction, rather than a specific direction of the variable trend. This test will determine the difference between the two answers “yes” or “no” based on the data (responses) collected in our survey. The number that we get from this test will be evaluated against the p-value when evaluating for success later. Additionally, different data visualization tools can be used to better understand the data. For example, a pie chart can be used to visualize the proportion of students who do and do not believe hotdogs are considered a sandwich.

Evaluation of Success:
A p-value < 0.05 indicates statistical significance, meaning that our main hypothesis was accurate. This means if our p-value is below 0.05, we can successfully say that the proportion of students at UVA who think a hot dog is a sandwich is not equal to 25%. However, if our p-value is greater than 0.05, we fail to reject the null hypothesis. This leads us to the conclusion that 25% of students at UVA think that a hot dog is a sandwich.